

Shyam Kumar Bhattarai

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PROFESSIONAL SUMMARY

I am a graduate student in Computer Science at Southeast Missouri State University with a 4.0 GPA, passionate about backend development and teaching. I have strong hands-on experience with Django and REST APIs and a solid understanding of data structures and algorithms. I am eager to apply my technical skills and teaching ability in a Graduate Assistant position, where I can contribute both to development projects and student learning.

EDUCATION

- Master of Science in Computer Science** **Expected Graduation: 2027**
Southeast Missouri State University, Cape Girardeau, MO *GPA: 4.0/4.0*
- Bachelor of Engineering in Information Technology** **Graduated: 2022**
Pokhara University, Nepal *GPA: 3.83/4.0 — Full Scholarship — Dean's List*

PROFESSIONAL EXPERIENCE

- Backend Developer (Django)** Nov 2023 – Jul 2025
SmartTech Nepal Pvt. Ltd., Nepal
 - Designed and maintained scalable backend systems using Django and Python, focusing on product features from planning through deployment.
 - Built and documented RESTful APIs, improving query response time by optimizing database performance.
 - Collaborated with frontend engineers to define API contracts and streamline integration, reducing iteration time and enhancing collaboration.
 - Refactored legacy code to improve maintainability, reduce technical debt, and ensure system stability.
- Computer Engineering Instructor** Nov 2022 – Oct 2023
Shree Shanti Secondary School, Nepal
 - Taught programming fundamentals, data structures, and computing concepts to secondary-level students, adapting lessons to different learning styles.
 - Designed and evaluated coursework and assessments, fostering student growth through hands-on projects and one-on-one guidance.
 - Helped improve class performance by offering personalized support to students struggling with programming concepts.
- C Programming Tutor (Part-Time)** 2021 – 2022
Nawtan Learning Institute, Nepal
 - Conducted theory and practical coding sessions for beginner-level students, teaching syntax, memory management, and debugging techniques.
 - Supported students in building confidence and developing problem-solving skills to tackle coding challenges independently.

PROJECTS

- 5G Network Coverage Analysis & Visualization**
MATLAB
 - Performed geospatial analysis to model signal propagation, identifying optimal cell tower placement for maximum coverage efficiency.
 - Created clear visual heatmaps in MATLAB to present findings to non-technical stakeholders.
- News Portal**
Django, PostgreSQL
 - Built a full-stack web application with user authentication, role-based access, and a content management system (CMS).
 - Managed dynamic article content and categories using relational database integration.
- Online Examination System**
Django, SQL
 - Developed a browser-based platform for timed exams, automatic grading of objective questions, and result reporting for instructors.
- Student Management System**
Django, SQL
 - Designed a system for tracking student records, attendance, and academic grades, with a user-friendly admin dashboard.

- Implemented a normalized relational schema with full CRUD operations and advanced search/filter functionality.
- **Task Management REST API**
Django REST Framework
 - Engineered a RESTful backend with JWT authentication, CRUD endpoints, and full API documentation using Postman.
- **Library Management System**
SQL
 - Designed a relational database schema and optimized queries for efficient record retrieval, including multi-table joins and aggregations.

TECHNICAL SKILLS

Languages:	Python, C, C++
Web & Backend:	Django, REST APIs, HTML, CSS, JavaScript
Databases:	MySQL, PostgreSQL
Tools:	Git, GitHub, MATLAB, Postman
CS Concepts:	Data Structures & Algorithms, Object-Oriented Programming, Software Engineering

ADDITIONAL STRENGTHS

- Strong verbal and written communication skills, with the ability to explain technical concepts to diverse audiences.
- Proven mentoring ability, having tutored and taught students across multiple years, enhancing their understanding of complex subjects.
- A collaborative team player who enjoys working with developers, students, and administrators to achieve common goals.